

Aman Srivastava

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EDUCATION

University of Maryland - College Park

August 2023 - May 2027

Bachelor of Science | Major in Computer Science | Minor in Philosophy

Relevant Coursework: Advanced Data Structures, Algorithms, OOP (Java), Discrete Structures, Programming Languages, Computer Systems, Data Science, Operating Systems, Computer and Network Security, Intro to AI

EXPERIENCE

IBM

May 2026-August 2026

Incoming Z Infrastructure Solutions Intern

Fintech Global Center

June 2025-August 2025

Software Engineering Intern

- Engineered a Python bond portfolio optimizer using SciPy Differential Evolution, maximizing yield under budget, duration/DV01, and credit rating constraints
- Designed a penalty-based objective function with feasible seeding, boosting convergence stability and ensuring regulatory-compliant asset weights
- Achieved 40% increase in efficiency for audit reports on portfolio metrics through robust automations.

Fintech Global Center

Software Engineering Intern

June 2024 - August 2024

- Reduced pricing error by 15% against benchmark models by developing a Python yield simulator
- Enabled identification of market reversal patterns for trading strategy backtesting by researching TensorFlow models leveraging RSI, Moving Averages, and Elliott Wave indicators
- Built a data preprocessing and visualization pipeline (NumPy, Matplotlib) enabling analysts to compare predictive vs actual yield curves interactively

ManTech

June 2022 - August 2022

Software Engineering Intern

- Developed a PowerApps audit platform integrated with SharePoint and Power Automate, automating security survey collection and data workflows across 100+ clients
- Consolidated cybersecurity assessment data into dashboards highlighting key vulnerabilities, generating actionable client reports and cutting manual review time by 50%

PROJECTS

Smart Meeting Summarizer App

- Built a full-stack meeting platform with React/TypeScript + Tailwind frontend and FastAPI backend, integrating OpenAI Whisper for transcription and GPT-4 for summarization
- Containerized backend with Docker Compose, PostgreSQL, MinIO for audio storage, and Redis for async processing across 8 REST API endpoints
- Implemented action item tracking with assignees, due dates, and status tracking, with CSV and email export

Autonomous AI Assistant

- Built an autonomous research agent with LangChain and OpenAI API that decomposes queries and generates JSON reports, cutting research time by ~60%
- Engineered a RAG pipeline with PDF ingestion (PyMuPDF), semantic chunking, and a Chroma vector database using OpenAI embeddings for sub-second similarity retrieval
- Integrated arXiv API for academic paper ingestion with end-to-end citation tracking and Pydantic validation for structured, hallucination-resistant outputs

Formula 1 Podium Predictor

- Developed an ML pipeline predicting F1 podiums using Random Forest and KNN models ($R^2 > 0.7$, 80% season accuracy) trained on Fast-F1 API telemetry
- Built an interactive D3.js frontend visualizing predicted vs. actual podium counts with season filtering, backed by a Python/Pandas data pipeline
- Engineered features from driver consistency, team performance, and historical podium rates with cross-validation and hyperparameter tuning

SKILLS

Languages: Java, Python, C, Rust, OCaml, SQL, JavaScript, TypeScript, HTML/CSS, x86 Assembly

Frameworks/Tools: FastAPI, Jupyter Notebook, React, Docker, PostgreSQL, Redis, Node.js, Git, Pandas, D3.js

Concepts: Data Structures & Algorithms, API Design & Integration, Full-stack, GenAI & LLM Integration

INTERESTS

Formula 1, Running, Hip Hop Dance, Bouldering, Watches, Cardistry, Financial Markets, Balisongs, Star Wars